

Multidimensional Goodhart

Response Channels, Scorecards, and Residual Shape

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Abstract. The initial hypothesis was that multidimensional proxy optimization would push residual error into more structured, perhaps more complex, hidden directions. The current framework replaces that with a conditional contract on what residual structure is licensed. Selection over fixed states licenses baseline response curves and δ -reweighting value-drift bounds. Intervention at fixed type requires action, cost, search, and stakes primitives. Multidimensional scorecards yield exchange-rate and entry effects rather than a generic “more metrics” sign: fixed-deficit per-agent harm is conserved across active measured sets if and only if harm-per-score satisfies $h_j = cw_j$. Residual shape is governed by these primitives together with the declared hidden value model. The transfer rule is a response-modeling contract, not a universal Goodhart theorem.

Motivation and positioning

Scalar Goodhart warnings compress two things: the target is reduced to a proxy, and the residual error is often reduced to an unnamed loss. In multidimensional settings the residual is the object of interest. It has direction, support, active constraints, exchange rates, and evidence requirements. The question is therefore not whether optimization can break proxies, but which calculation about residual structure is licensed once the response channel, action/search geometry, aggregation rule, and hidden value model are declared.

Let $s \in S$ denote the latent state of a unit (an agent, a type, or a production realization), with baseline law μ on $(S, \mathcal{F})$. Write $G(s) \in \mathbb{R}^m$ for target-relevant state, $P(s) \in \mathbb{R}^k$ for proxy features, φ for the intended proxy relation, and $\varepsilon(s) = P(s) - \varphi(G(s))$ for proxy residual. A claim about multidimensional Goodhart must say how optimization pressure changes the law of s (selection) or the behavior of fixed types (intervention).

This refines Manheim and Garrabrant’s regressional/extremal/causal/adversarial taxonomy by mapping the variants onto channel-specific licensed calculations, rather than treating the taxonomy itself as the formal object: their selection variants correspond to the reweighting channel, causal and adversarial Goodhart to the intervention channel with declared cost and worst-case primitives. The scorecard analysis extends Holmstrom–Milgrom multitasking to a declared hidden-harm vector, recovering the substitution geometry and adding an exchange-rate conservation criterion that is absent from the LEN benchmark.

Contributions

- **Channel separation.** Selection (reweighting at fixed kernel) and intervention (kernel change at fixed type) are formally distinct channels with distinct licensed calculations.
- **Selection bounds.** δ -reweighting bounds for hidden-coordinate drift and for any declared scalar value $v \cdot H$; the coordinate norm and value vector are declared inputs, not inferred from proxy movement.
- **Exchange-rate conservation.** Fixed-deficit per-agent harm $H_{\text{per}(M,d)}$ is conserved across active measured sets if and only if $h_j = cw_j$, the scorecard analogue of, and extension to, Holmstrom–Milgrom multitasking.
- **Adaptive-hardening boundary.** Under the additive, separable-quadratic, fixed- (M, d, V) contract, fixed-deficit gaming stops exactly when $S_{t(M)} < \frac{d^2}{2V}$.
- **Response-modeling contract.** The deflationary framework itself: each applicable calculation requires a declared set of primitives, and unlicensed transfers are named as such.

Current Findings

- **Selection response.** A pure selection channel reweights a fixed baseline law: $\mu_\theta = L_\theta \mu$. Hidden drift is $B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu[H]}$. Threshold response and Boltzmann response are baseline path functionals; covariance is only a local velocity where the tilted moment-generating function is finite.
- **Selection/value bounds.** If $\delta = \|L_\theta - 1\|_{L^2(\mu)}$ and hidden coordinates have standard deviations s_i , then $|B_{H_i}| \leq \delta s_i$ and $\|B_H\|_2 \leq \delta \|s\|_2$ after declaring the coordinate norm. For a declared scalar value vector v , $|\Delta(v \cdot H)| \leq \delta \sqrt{v^T \Sigma_H v}$. The value metric is a modeling input, not inferred from proxy movement.
- **Intervention channels.** Intervention means the response kernel changes at fixed type: $K_{\theta(ds|u)} \neq K_0(ds|u)$ on a positive-mass set. The one-dimensional quadratic threshold toy gives a gaming wedge $\Delta = \sqrt{2\kappa V}$; this is a signature of ease of gaming and stakes, not a baseline covariance result. For convex action cost c and score deficit d , the private score-deficit cost is $m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)]$. This is not a welfare bound without a hidden-harm functional.
- **Scorecards: the exchange-rate criterion.** The cleanest empirical-design hook in the framework. With additive score, separable quadratic costs, fixed deficit d , score weights w_j , and hidden harm rates h_j , fixed-deficit per-agent harm is $H_{\text{per}(M,d)} = d \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}$. Conservation across active measured sets requires $h_j = cw_j$, the invariance criterion that tells designers what to measure to know whether adding a metric helped, hurt, or merely re-routed harm. Population harm is a separate entry object; conjunctive aggregation changes the comparison again.
- **Response shape.** Proxy pressure does not generically select minimum-complexity residuals. Quadratic costs select cost-minimal directions; fixed charges and caps can create lumpy active-set switches; low-rank affordances constrain

drift to an image; search-prior claims require a predeclared coding or search process, so they are not free-floating simplicity statements but contracts on the description language.

Why the Framework Matters

The framework replaces slogan transfer with primitive transfer. A domain application must declare type space U , baseline behavior, policy exposure, selection weights W_θ and/or response kernels K_θ , action/search geometry, proxy/target relation, aggregation, hidden harm/value, and evidence standard. The same score movement can be selection, harmful fixed-type gaming, harmless proxy repair, real improvement, or a mixture. Without the contract, a Goodhart explanation is often just a retrospective label.

This matters most when importing formal analogues. Lucas-style policy non-invariance, strategic classification, multitask incentives, adaptive holdout reuse, reward-model overoptimization, and benchmark contamination are nearby warnings, but each supplies only some primitives. None by itself gives a hidden welfare model, an action-cost geometry, an empirical estimate of κ , or a scorecard exchange rate. The contract says which calculation is licensed: a selection response curve, a reweighting bound, an intervention budget, an aggregation comparison, or a shape prediction.

The framework is also intentionally lossy about policy. It can say that a toy diagnostic is unavailable when effective weights, hidden harm, response ease, or stakes are not declared. It can say that aggregate score movement is insufficient evidence for hidden-target improvement. It cannot rank policies or infer welfare from score movement without additional empirical and value commitments.

Live Boundaries and Next Work

- The deterministic adaptive-hardening boundary is closed only in the finite-channel additive-score, separable-quadratic, fixed- M , fixed- d , fixed- V , fixed-weight, deterministic-observation contract: fixed-deficit gaming stops exactly when $S_{t(M)} = \sum_j \kappa_{j,t} w_j^2 < \frac{d^2}{2V}$.
- Stochastic observation, repair-budget design, endogenous stakes, changing measured sets or deficits, shared bottlenecks, nonconvex geometry, and policy optimization remain separate dynamic problems.
- Empirical applications require repeated-type observations, exposure variation, action traces, cost variation, or predeclared hidden outcomes. The framework does not license hidden-target estimates, welfare or research-value inference, RLHF claims, or generic advice that more metrics are good or bad without those primitives.
- **Anti-application and falsification.** The framework is most useful where primitives are declarable and reasonably stable. It fails to be useful when type space drifts, measured sets are unobservable, or no defensible hidden value model is available; a framework that infers welfare from score movement alone does something this contract refuses to do, and that refusal is a limitation when stakeholders need a verdict without primitives. The contract itself would be falsified by a domain where licensed calculations systematically disagree with outcomes that primitive declarations were supposed to track.